AIN SHAMS UNIVERSITY
Faculty of Engineering
Electronics and Communications Engineering Department

Course Code: ECE XXX
Course Title: Self Learning

Fabry-Pérot Etalon Project Documentation

Technical Simulation & Analysis

Author:
Mohammed Nader
Student ID: 12345678

Submitted To:
Dr. First Last
Course Instructor

July 2026

0. Contents

1. Introduction

A Fabry Perot Etalon(also known as an optical cavity) is a device made of two mirrors separated by a distance L , when light enters the etalon , it undergoes certain phenomena , that either makes it interfere constructively or destructively based on the parameters of the etalon itself , and the wavelength of the light , making it act as a wavelength selector

2. Theory

Since the mirrors on either side of the cavity, are not fully reflective , light that enters the cavity will eventually Transmit out of it , the equation of the transmittance is as follows

$$T(\lambda) = \frac{1}{1 + F\sin^2\left(\frac{\delta}{2}\right)} \tag{1}$$

where the phase shift δ

$$\delta = \frac{4\pi n L}{\lambda}$$

and the finesse coefficient F

$$F = \frac{4R}{(1-R)^2}$$

Notice that the transmission coefficient $T(\lambda) = 1$, when $sin^2\left(\frac{\delta}{2}\right) = 0$, which occurs at $\delta = 2m\pi$, this gives the resonance frequencies of the Fabry Perot etalon , notice that those frequencies are a function of n and L , which are intrinsically device parameters Plugging in $\delta = 2m\pi$ gives the resonant wavelengths

$$\lambda_m = 2nL/m$$

and thus , the resonant frequencies are

$$\nu_m = cm/2nL$$

where m is a positive integer and c is the speed of light in air
Another very important parameter that stems from the resonant frequencies' equation , is the spectral distance between each resonant frequency and the next , also known as the Free Spectral Range(FSR)

$$\nu_{FSR} = c/2nL$$

As seen in the equation , a longer cavity results in the resonant peaks of transmission being closer together , whereas a smaller cavity results in wider spaced peaks

The final parameter of interest for this documentation is the Finesse , a dimensionless quantity that characterizes the sharpness of the cavity's resonances, given by the equation:

$$\mathscr{F} = \frac{\pi\sqrt{R}}{1-R}$$

As R increases, that is, the mirrors are less lossy, the finesse $\mathscr{F}$ increases too, indicating sharper peaks it is also sometimes referred to as the Quality factor of the optical cavity

3. Effect of Reflectivity on Transmission Spectrum

For the first part , we shall examine how sweeping through 3 different values of Reflectivity $R = \{0.5, 0.8.0.95\}$ affects the transmission coefficient $T(\lambda)$

Code snippet:

```
    L=10e-6 # cavity length
Reflectivety_values=[0.5,0.8,0.95]
plt.figure(figsize=(15,4))
for R_1 in Reflectivety_values:
    F_loop,T_Loop= compute_transmission(R=R_1, L=L,n=n , wavelength=lambda_m)
```

where `compute_transmission` is simply a plug in function that evaluates (1)

Which Results in the following Plot:

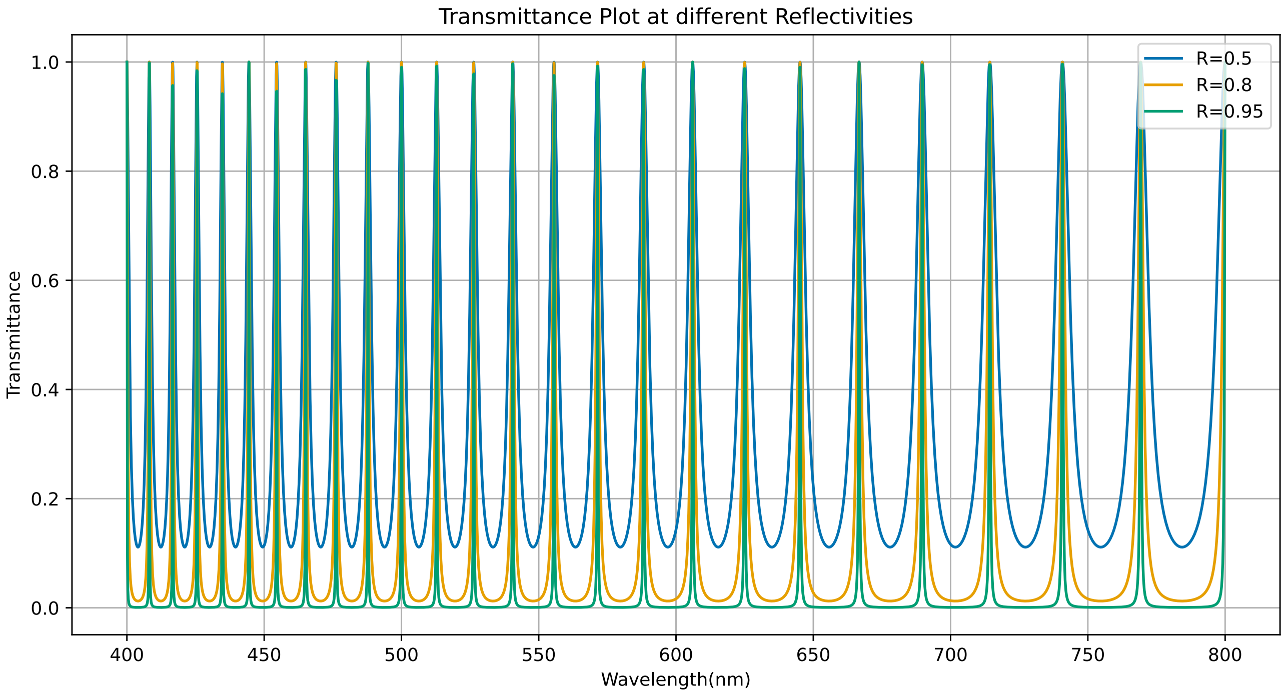

Figure 1: Transmission spectrum across different reflectivities R ($L = 10\ \mu$m)

The effect shown in the figure ,is that as R increases the peaks become much spectrally sharper because more light becomes trapped in the cavity causing more interference which narrows the transition bands.

When R is low, most light escapes the cavity after just a few reflections. When R is high , light reflects back and forth hundreds of times before finally escaping. The number of interfering beams effectively multiplies, trapping the light longer

4. Effect of Cavity Length on Free Spectral Range

In the introduction section , it was mentioned that by inspection of the FSR equation , there is an inverse Relation between the Length of the optical cavity L and the spectral distance between the resonance peaks , lets examine that relation over the set of values $L = \{5\mu m, 10\mu m, 20\mu m\}$

Code snippet:

```
    L_values=np.array([5,10,20])*1e-6
plt.figure(figsize=(15,4))
for L_1 in L_values:
    _,T_Loop2= compute_transmission(R=0.9,L=L_1,n=n,wavelength=lambda_m)
```

Which Outputs the Plot:

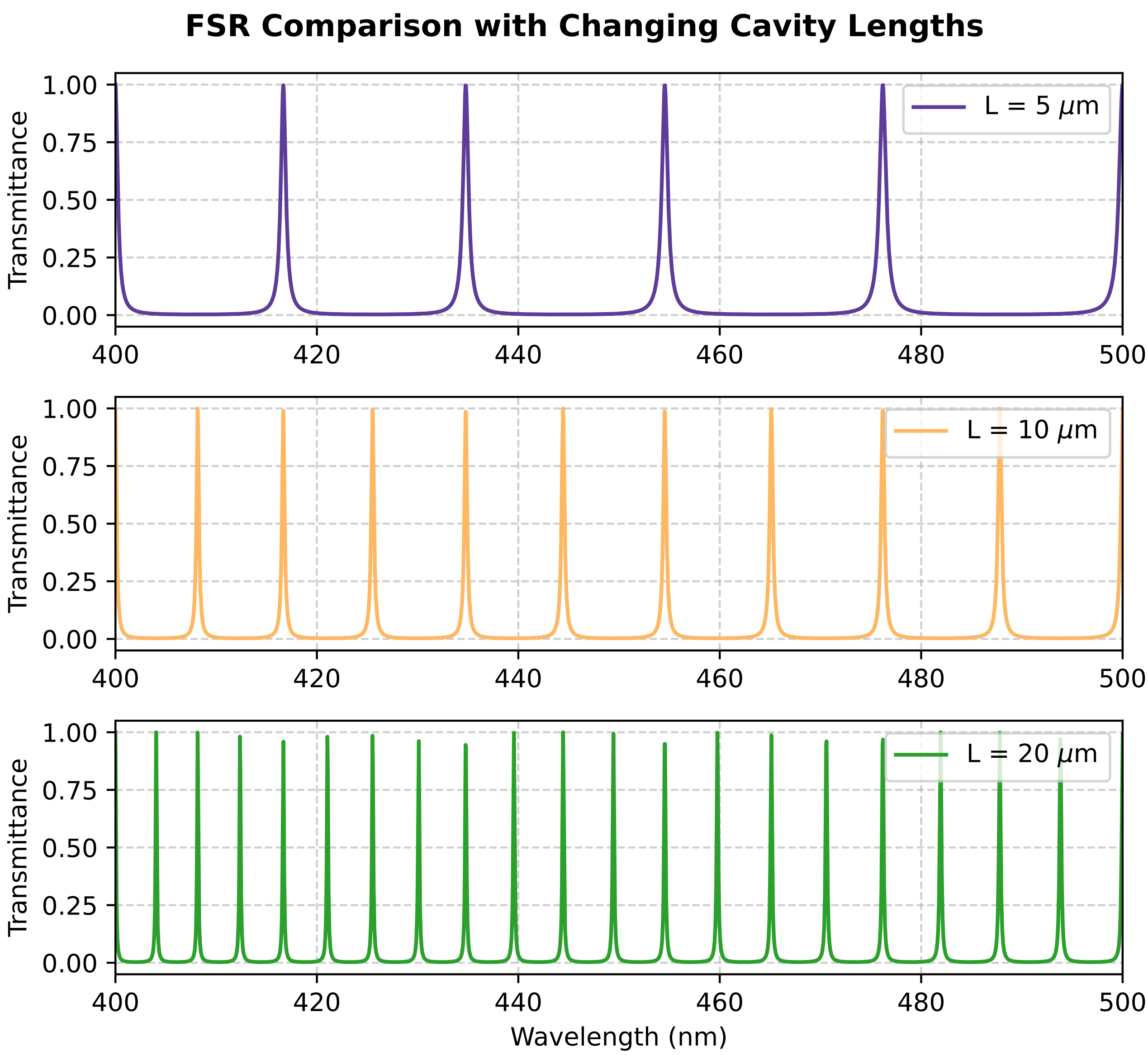

Figure 2: FSR Comparison across different Cavity Lengths L

As L increases the peaks are much closer in spectral distance together , which aligns with what said in the introduction section.

5. Finesse Comparison: 2 Methods

An equation for the finesse of an optical cavity $\mathscr{F}$ can be found in the introduction section , however finesse is defined as how many resonance linewidths fit into the distance between adjacent peaks, which is the equation

$$\mathscr{F} = FSR/FWHM$$

where FSR and FWHM can either both be measured in m or in hz , and the FWHM is the spectral distance between the two points at which the peak reached half of its maximum value.

This section aims at comparing the two formulas and seeing the error between them .

We shall call the definition $\mathscr{F}_{measured}$ and the formula given in the introduction $\mathscr{F}_{theoritical}$ where

$$\mathscr{F}_{theoritical} = \frac{\pi\sqrt{R}}{1-R}$$

5.1 Theoretical Finesse

```
    R_array= np.array(Reflectivety_values)
Finesse_values_equation = (np.pi * np.sqrt(R_array)) / (1 - R_array)
```

Code Output [4.44288294 ,14.04962946, 61.24091502]

5.2 Measured Finesse

```
for R_1 in Reflectivety_values:
    _, T_Loop = compute_transmission(R=R_1, L=L, n=n, wavelength=lambda_m)
    peaks, _ = find_peaks(T_Loop)

    finesse, fsr, fwhm = calculate_measured_finesse(lambda_m, T_Loop, peaks)
       # helper function

    #Check appendix for full method used to calculate fwhm and fsr
```

Code Output [4.46,14.41.62.94]

Comparison Table:

Reflectivity (R)	**Theoretical Finesse**	**Measured Finesse**	**Percent Error (%)**
0.50	4.44	4.46	0.45%
0.80	14.05	14.41	2.56%
0.95	61.24	62.94	2.78%

Table 1: Comparison of Theoretical and Measured Finesse Values

Although both measurements are not exactly equal , the error is minimal , which proves both methods are effective in measuring the finesse $\mathscr{F}$ of the cavity.

6. Conclusion

This project examined the nature of an air filled Fabry Perot etalon and how changing its parameters affects certain Physical quantities.
The fundamental property of a Fabry Perot etalon is that it transmits light at specific resonant frequencies, which makes it useful in WDM telecommunication channels.

7. Appendix

```
    import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import find_peaks
import os
#parameters and function defintion
n=1 # air
c=3e8 # speed of light in air
lambda_m =np.linspace(400,800,10000000) * 1e-9 #10,000 points between 400nm
    and 800nm
wavelength_step_nm = (lambda_m[1] - lambda_m[0]) * 1e9
T_by_R = {}
Peaks_by_R = {}
def compute_transmission(R,L,n,wavelength):
    delta_m= 4*np.pi*L*n/wavelength
    F=(4*R)/(1-R)**2
    T= 1/(1+F*(np.sin(delta_m/2)**2))
    return F,T
L=10e-6 # cavity length
Reflectivety_values=[0.5,0.8,0.95]
plt.figure(figsize=(15,4))
for R_1 in Reflectivety_values:
    F_loop,T_Loop= compute_transmission(R=R_1, L=L,n=n , wavelength=lambda_m)
    plt.plot(lambda_m*1e9,T_Loop,label=f'R={R_1}')
    #Part 3 , To get finesse from graph
    T_by_R[R_1] = T_Loop
# ok now i have 3 vectors representing each R , to find FSR and fwhm for each
    Peaks_Loop,_= find_peaks(T_by_R[R_1])
    Peaks_by_R[R_1]=Peaks_Loop
    mid_idx = len(Peaks_Loop) // 2
    middle_peak_idx = Peaks_Loop[mid_idx]
    next_peak_idx = Peaks_Loop[mid_idx + 1]
    FSR = (next_peak_idx - middle_peak_idx) * wavelength_step_nm

    # Slice a local window
    FSR_indices = next_peak_idx - middle_peak_idx
    window_radius = FSR_indices // 2
    window_start = middle_peak_idx - window_radius
    window_end = middle_peak_idx + window_radius
    T_window = T_Loop[window_start:window_end]

    indices_above_half = np.where(T_window >= 0.5)[0]

    # index width
    fwhm_index_width = indices_above_half[-1] - indices_above_half[0]

    FWHM = fwhm_index_width * wavelength_step_nm

    # Calcuate Finesse
    measured_finesse = FSR / FWHM
```

```
    print(f"R={R_1} ,F={measured_finesse}")
plt.title('Transmittance Plot at different Reflectivities')
plt.xlabel('Wavelength(nm)')
plt.ylabel('Transmittance')
plt.grid()
plt.legend(loc="upper right")
#Part 2
L_values=np.array([5,10,20])*1e-6
plt.figure(figsize=(15,4))
plt.gca().set_prop_cycle(color=['#5E3C99', '#FDB863', '#2CA02C'])
for L_1 in L_values:
    _,T_Loop2= compute_transmission(R=0.9,L=L_1,n=n,wavelength=lambda_m)
    plt.plot(lambda_m*1e9,T_Loop2,label=f'L={np.ceil(L_1/1e-6)}µm')
plt.title('FSR comparison with changing cavity Lengths')
plt.xlabel('Wavelength(nm)')
plt.ylabel('Transmittance')
plt.grid()
plt.legend(loc="upper right")
plt.xlim(400,500)
    #PART 3
# Calculate F from each R
R_array= np.array(Reflectivety_values)
Finesse_values_equation = (np.pi * np.sqrt(R_array)) / (1 - R_array)
print(f"R= 0.5, F={Finesse_values_equation[0]}")
print(f"R=0.8 , F={Finesse_values_equation[1]}")
print(f"R=0.95 , F={Finesse_values_equation[2]}")
plt.show()
```